

Predictors of Hospital Onset Infection: A Matched Retrospective Cohort Dataset

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Abstract

This repository contains a de-identified and curated patient-level dataset for modeling the impact of fine-grained environmental and patient-level factors on nosocomial acquisition of a wide range of drug-susceptible and drug-resistant pathogens. The data are derived from the clinical data warehouses of the Mass General Brigham health system and span all adult inpatients admitted between May 2015 and July 2024.

We use a matched case/control design and provide two datasets reflecting distinctive but complementary strategies to model our outcome. The first, which we call the **environmental analysis**, investigates the impact of the hospital unit microbiome, using colonization pressure as a proxy, and matches on prior antibiotic exposure to equalize its selective impact across cases and controls. The second strategy, which we call the **patient analysis**, investigates the impact of prior fine-grained antibiotic exposures and matches controls spatiotemporally to cases based on their time of entry into the same ward, to equalize the impact of the hospital microbiome.

Each observation in the dataset corresponds to a hospitalization. We look at nosocomial acquisition of 11 common drug-susceptible and drug-resistant pathogens, with cases having a culture positive for the target pathogen between day 3 and day 30 after admission into their room.

Background

Nosocomial acquisition of pathogens is a common and serious complication of medical care. On any given day in the United States, approximately one in 31 hospitalized patients has at least one healthcare-associated infection, underscoring the clinical and public health importance of robust prediction and prevention frameworks [1].

Prior research has shown that both environmental and patient-specific factors are important predictors. From an infection control epidemiology standpoint, two recurring drivers are (i) ward-level exposure, captured by constructs such as colonization pressure [2] and prior-room occupant effects, and (ii) patient-level exposure, including recent and cumulative antibiotic use [3–5]. For example, Gu et al. (2023) showed that prior bed occupants colonized or infected with multidrug-resistant organisms significantly increased the risk of acquisition in subsequent occupants, supporting the notion that microenvironment-level pressure matters [5]. Similarly, Jolivet (2020) showed that colonization pressure in the previous week was associated with acquisition of extended-spectrum beta-lactamase-producing Enterobacterales in medical and surgical intensive care units, but not for MRSA [6].

However, the results of these studies have been difficult to operationalize, as they lack the granularity to inform decisions for patients who have a multitude of risk factors. Furthermore, the absence of publicly available data and open-source code makes reconstructing these models infeasible. We seek to fill this gap by making available two datasets that approach the problem of predicting hospital-acquired infection from electronic health record (EHR) data in unique but complementary ways. This dual-matched design, with one cohort matched by antibiotic exposure and the other by hospital unit, enables a separation of environmental and pharmacologic risk factors rarely possible in prior resources. We also publish a highly annotated code base of an end-to-end pipeline for reproducing this data locally. Together, these datasets represent one of the largest publicly available EHR datasets published for investigating hospital-acquired infection, covering 11 target pathogens and estimating risk from 14 different antibiotic classes and 9 different colonization pressure groups. The scale, high level of curation, and code base make this dataset uniquely suited for both mechanistic and machine learning studies of nosocomial infection risk.

The **environmental analysis** seeks to estimate the impact of a hospital unit's microbiome on the risk of hospital-acquired infection by matching controls to cases across 14 different classes of antibiotics. By doing so, we equalize the selective pressure of antibiotic exposure across cases and controls. Given the finite number of indications for antibiotics, it also serves to equalize a number of other confounders, such as comorbidity burden. Our primary predictor is colonization pressure, which represents the prevalence of an organism (or set of organisms) in ward co-occupants at the time of entry by the case or control.

The **patient analysis** is the converse of the environmental analysis and seeks to understand the impact of exposure to a given antibiotic class on the risk of hospital-acquired infection. Consistent with recent cohort data on hospital-onset *Clostridioides difficile* infection, antibiotic exposure is a key modifiable predictor of risk, supporting the need to model class-specific exposure histories [4]. Here, controls are selected to be on the same hospital unit in the same month and year as cases, thereby eliminating differences that may occur due to exposure to a unit's microbiome. The primary predictors are prior exposure to a given antibiotic class.

For both analyses, we apply a number of other criteria to ensure both cases and controls did not have prior evidence of infection and were not being actively treated for infection before the observation period. Controls were also matched on demographics, prior surgery, and length of stay in the room.

Methods

Study design

The analysis consists of a matched case-control design [7, 8].

Cohort Selection

Cases and controls were identified based on their room occupancy status, microbiology results, and antibiotic exposure histories.

Time zero (T_0 , where the subscript represents days) was defined as the timestamp of entry into the qualifying room. This room always represented the first room in a patient's hospital stay lasting >48 hours, excluding any time in the emergency department.

Inclusion Criteria

- **Single qualifying room stay:** Patients must have stayed in only one room for >48 hours during the observation period (T_{+3} to T_{+30}). They may be discharged from their room prior to the end of the observation period.
- **No clinical evidence of active infection at the time of admission:** Patients must not have received any antibiotics from T_{-7} to T_{+3} .
- **No clinical evidence of prior colonization:** No clinical or surveillance cultures positive for any target organism in the previous 6 months (or 12 months if the organism was *S. aureus* or an *Enterococcus* species).

Exclusion Criteria

- Admissions limited to emergency room visits or with no ward stay >48 hours in duration.
- Admissions where the patient stayed in >1 room for >48 hours.
- Any antibiotic exposure between day -7 and day $+3$.
- Positive clinical or surveillance culture with a target organism within 6 months prior to T_0 (12 months for *S. aureus* or *Enterococcus* species).
- Missing demographic data (age, sex) or invalid date/time stamps preventing accurate assignment of relative days.

Case Definition

- Cases were patients with identification of a target organism (defined below) in any clinical or surveillance culture between T_{+3} and T_{+30} , regardless of whether discharge occurred before day 30.

Control Definition

- Controls were patients who met inclusion criteria but did not have evidence of the target organism in any clinical or surveillance culture between T₊₃ and T₊₃₀. Matching strategy described below.

Matching Strategy

For both analyses, controls were matched to cases using predefined demographic, clinical, and temporal criteria to ensure comparable exposure windows and infection risk contexts.

A maximum of three controls were selected per case (1:≤3 matching ratio), depending on the number of eligible controls within the cohort. A single control could be matched to multiple cases if it satisfied all matching criteria, allowing reuse across case sets to preserve cohort representativeness and maximize matching completeness.

Environmental analysis

- Controls were matched to cases based on:
 - Demographics (age ±5 years, sex)
 - Prior surgery (of any type) in the previous 90 days (1/0)
 - Length of stay in room (±3 days) or length of time to case's index culture (whichever was shorter)
 - Antibiotic exposure, stratified by:
 - Number of courses (defined as ≥2 days of an antibiotic, ≥7 days apart from a different course)
 - Antibiotic class (and corresponding members):
 - Anti-anaerobic
 - Metronidazole
 - Penicillins
 - Penicillin, amoxicillin, ampicillin
 - Extended-spectrum penicillins
 - Amoxicillin-clavulanate, ampicillin-sulbactam, piperacillin-tazobactam
 - Anti-staphylococcal penicillins
 - Dicloxacillin, nafcillin, oxacillin, cefazolin
 - Cephalosporins
 - Cephalexin, cefadroxil, cefaclor, cefuroxime, cefprozil, cefpodoxime, cefdinir, cefotetan, cefixime, cefoxitin
 - Extended-spectrum cephalosporins
 - Cefotaxime, ceftazidime, ceftriaxone, cefepime, ceftaroline, cefiderocol
 - Carbapenems
 - Ertapenem, imipenem, meropenem
 - Fluoroquinolones
 - Moxifloxacin, ciprofloxacin, levofloxacin, delafloxacin
 - Folate inhibitors
 - Sulfamethoxazole, trimethoprim, trimethoprim-sulfamethoxazole
 - Glycopeptides
 - Systemic: vancomycin
 - Anti-*C. difficile*: vancomycin (oral)
 - Lincosamides
 - Clindamycin
 - Macrolides
 - Azithromycin, clarithromycin
 - Tetracyclines
 - Tetracycline, demeclocycline, doxycycline, minocycline, tigecycline, omadacycline, eravacycline

Patient analysis

- Controls were matched to cases based on:
 - Hospital ward
 - Admission month and year
 - Length of stay in room (±3 days) or length of time to case's index culture (whichever was shorter)

Outcomes

Nosocomial acquisition (T₊₃ to T₊₃₀) of one of the following target pathogens identified in any clinical or surveillance culture:

- *Escherichia coli* (susceptible to β-lactams)
- Extended-spectrum β-lactamase-producing (ESBL) *E. coli*
- *Klebsiella pneumoniae* (susceptible to β-lactams)

- Extended-spectrum β -lactamase-producing (ESBL) *K. pneumoniae*
- *Clostridioides difficile*
- Vancomycin-susceptible *Enterococcus faecalis*
- Vancomycin-resistant *E. faecium*
- Methicillin-susceptible *Staphylococcus aureus* (MSSA)
- Methicillin-resistant *S. aureus* (MRSA)
- Drug-susceptible *Pseudomonas aeruginosa* (susceptible to ceftazidime, cefepime, piperacillin-tazobactam, and carbapenems)
- Drug-resistant *P. aeruginosa* (resistant to at least one β -lactam)

Antibiotic susceptibility phenotypes were derived from the 32nd edition of the M100 document, published by the Clinical and Laboratory Standards Institute.

Model covariates

Environmental analysis

- Colonization pressure, defined as the time-weighted prevalence of a specified organism, or an organism set, in ward co-occupants, given by:

$$\text{Study participant}_{\text{ward}_i} = \sum_{\text{Co-occupant}_1}^{\text{Co-occupant}_j} Y_j \cdot e^{(-\lambda \cdot T_j)}$$

- A ward co-occupant was defined as a person with a >48-hour stay in the 30-day period prior to entry into the ward by the study participant (T_{-30} to T_0).
- $Y_j = 1$ if the ward co-occupant had the target organism within the past 6 months; $Y_j = 0$ if not.
- T_j is the time interval in days between the date the organism was identified in the co-occupant and the date of entry into the ward by the study participant (case or control).
- λ is a tunable decay parameter controlling the influence of time (set to 0.01 by default); this accounts for the diminishing effect of prior organisms over time.
- The look-back period for Y_j and T_j was set to 6 months to account for the possibility of organism carriage.
- CP was calculated separately for the following nine organisms and groups (drug-susceptible, drug-resistant, and ESBL phenotypes are defined as above for the target organisms):
 - Drug-susceptible Enterobacterales
 - ESBL Enterobacterales
 - Vancomycin-susceptible *Enterococcus* (VSE) species
 - Vancomycin-resistant *Enterococcus* (VRE) species
 - *C. difficile*
 - MSSA
 - MRSA
 - Drug-susceptible *P. aeruginosa*
 - Drug-resistant *P. aeruginosa*
- Example of use:
 - Assume patient_x enters ward_A on T_0 , and there are five patients who resided on ward_A between T_{-30} and T_0 who had *C. difficile* in the previous 6 months, at a median of 60 days prior to T_0 .
 - The corresponding colonization pressure for *C. difficile* on ward_A for the day patient_x is admitted would therefore be 2.744.
- Comorbidities defined by the Elixhauser Index [9]

Patient analysis

- Age
- Sex
- Comorbidities defined by the Elixhauser Index [9]
- Number of courses of the following antibiotic classes in the previous 90 days (members of each class defined above):
 - Anti-anaerobic
 - Penicillins
 - Extended-spectrum penicillins
 - Anti-staphylococcal penicillins
 - Cephalosporins
 - Extended-spectrum cephalosporins
 - Carbapenems
 - Fluoroquinolones
 - Folate inhibitors

- Glycopeptides
- Lincosamides
- Macrolides
- Tetracyclines

Data Description

Note that the dataset `H0_infxn_analysis.csv` contains data for both the environmental and patient analyses.

Sample Sizes

Environmental analysis

- 14,923 cases were matched to 28,480 control entries for 11 pathogens. The same control can be matched to different pathogens during the matching process.

Sample sizes for environmental analysis cohorts

Pathogen	Case	Control
<i>E. coli</i>	3,585	5,693
ESBL <i>E. coli</i>	996	2,299
<i>K. pneumoniae</i>	1,704	3,684
ESBL <i>K. pneumoniae</i>	238	376
Vancomycin-susceptible <i>E. faecalis</i>	1,823	3,082
Vancomycin-resistant <i>E. faecium</i>	244	366
<i>C. difficile</i>	592	881
MSSA	2,956	6,504
MRSA	1,101	2,397
Drug-susceptible <i>P. aeruginosa</i>	1,309	2,533
Drug-resistant <i>P. aeruginosa</i>	379	665

Patient analysis

- 16,933 cases were matched to 39,372 control entries for 11 pathogens. The same control can be matched to different pathogens during the matching process.

Sample sizes for patient analysis cohorts

Pathogen	Case	Control
<i>E. coli</i>	4,576	9,875
ESBL <i>E. coli</i>	1,034	2,511
<i>K. pneumoniae</i>	1,723	4,194
ESBL <i>K. pneumoniae</i>	336	804
Vancomycin-susceptible <i>E. faecalis</i>	2,384	5,698
Vancomycin-resistant <i>E. faecium</i>	389	962
<i>C. difficile</i>	1,094	2,558
MSSA	2,375	5,387
MRSA	1,102	2,656
Drug-susceptible <i>P. aeruginosa</i>	1,456	3,615
Drug-resistant <i>P. aeruginosa</i>	464	1,112

Feature Dictionary

- `analysis`
 - Description: Matching protocol
 - Type: Categorical
 - Encoding: N/A

- Units: N/A
- Range: environmental / patient. Descriptions of both analyses in Methods
- **run**
 - Description: The target pathogen
 - Type: Categorical
 - Encoding: N/A
 - Units: N/A
 - Range: The 11 pathogen targets described in the Methods section
- **group**
 - Description: Binary indicator of whether the entry is a case or a control
 - Type: Categorical
 - Encoding: N/A
 - Units: N/A
 - Range: Case / Control
- **group_binary**
 - Description: Binary indicator of whether the entry is a case or a control
 - Type: Categorical
 - Encoding: 1: Case / 0: Control
 - Units: N/A
 - Range: 1 / 0
- **group_index**
 - Description: Index to identify cases and controls in the same group
 - Type: Categorical
 - Encoding: 1: Case, 0: Control
 - Units: N/A
 - Range: 1 / 0
- **deidentified_patient_id**
 - Description: De-identified patient ID (see **Data De-identification**)
 - Type: Integer
 - Encoding: N/A
 - Units: N/A
 - Range: Environmental analysis: 1–34,338; Patient analysis: 1–49,583
- **deidentified_month**
 - Description: De-identified month of entry into a hospital unit; dates internally consistent for a patient (see **Data De-identification**)
 - Type: Integer
 - Encoding: N/A
 - Units: N/A
 - Range: 1–12
- **deidentified_year**
 - Description: De-identified year of entry into a hospital unit; dates internally consistent for a patient (see **Data De-identification**)
 - Type: Integer
 - Encoding: N/A
 - Units: N/A
 - Range: 2000–2100
- **dept_code**
 - Description: De-identified code for hospital ward (see **Data De-identification**)
 - Type: Integer
 - Encoding: N/A
 - Units: N/A
 - Range: 1–344
- **duration**
 - Description: Length of stay in the first room
 - Type: Numerical
 - Encoding: N/A
 - Units: days
 - Range: 2–30
- **time_to_infxn**
 - Description: Time between entry into the room (T_0) and the date of positive culture with the pathogen of interest. Applies to cases only; values for controls are set to N/A.
 - Type: Numerical
 - Encoding: N/A

- Units: days
 - Range: 3.1–30 / N/A
- **matching_duration**
 - Description: Length of time for matching controls to cases; ± 3 days of the length of stay (**duration**) or the timestamp for index culture (**time_to_infxn**), whichever is shorter.
 - Type: Numerical
 - Encoding: N/A
 - Units: days
 - Range: 3.1–30
- **age**
 - Description: Age of the patient at T_0 ; patients ≥ 90 are grouped into a single category
 - Type: Numerical
 - Encoding: N/A
 - Units: years
 - Range: 0–90 / >90
- **sex**
 - Description: Sex of the patient
 - Type: Categorical
 - Encoding: N/A
 - Units: N/A
 - Range: Male / Female
- **any_surgery**
 - Description: Any prior surgery in the previous 90 days
 - Type: Binary
 - Encoding: 1: Had prior surgery, 0: No prior surgery
 - Units: N/A
 - Range: 0 / 1
- **any_abx_0_60**
 - Description: Binary indicator if the patient received any antibiotics in the previous 60 days
 - Type: Binary
 - Encoding: 1: Had prior antibiotic exposure, 0: No prior antibiotic exposure
 - Units: N/A
 - Range: 0 / 1
- **{abx_name}_0_60**
 - Description: Number of courses of **{abx_name}** in the previous 60 days, where each course is at least 2 days in duration
 - Type: Integer
 - Encoding: N/A
 - Units: Number of courses
 - Range: **{abx_name}**-specific ranges
- **{pathogen_name}_cp**
 - Description: Ward-level colonization pressure of **{pathogen_name}**. Calculated by the time-weighted 6-month prevalence of **{pathogen_name}** in ward co-occupants who occupied the same ward as a case or control in the previous 30 days
 - Type: Numerical
 - Encoding: N/A
 - Units: N/A
 - Range: **{pathogen_name}**-specific ranges
- **elix_index_mortality**
 - Description: Elixhauser Index calculated based on [9]
 - Type: Numerical
 - Encoding: N/A
 - Units: N/A
 - Range: Environmental: –35–101; Patient: –33–101
- **elix_{elix_name}**
 - Description: Indicator of individual Elixhauser category
 - Type: Binary
 - Encoding: 1: Specific Elixhauser comorbidity (**{elix_name}**) present, 0: Specific Elixhauser comorbidity (**{elix_name}**) absent
 - Units: N/A
 - Range: 0 / 1

Dependent Files (not included)

- `micro.csv`
 - Description: Microbiology culture information
 - Features in final dataset derived from this dataset:
 - `pathogen` – Pathogen ID
 - `antibiotics` – Antibiotic used for susceptibility testing
 - `lab_result` – Susceptibility testing result
- `dems.csv`
 - Description: Demographic information
 - Features in final dataset derived from this dataset:
 - `age` – Age of the patient at T₀
 - `sex` – Sex of the patient
- `ADT.csv`
 - Description: Admission, discharge, and transfer timestamps for all hospital encounters
 - Features in final dataset derived from this dataset:
 - `admission_date` – Admission timestamp for the patient
 - `admission_loc` – Admission location or source of the patient
- `abx.csv`
 - Description: Antibiotic prescriptions
 - Features in final dataset derived from this dataset:
 - `any_abx_0_60` – Whether the patient was treated with antibiotics within 60 days
 - `{abx_name}_0_60` – Number of courses of `{abx_name}` in the previous 60 days (refers to antibiotic class)
- `elix.csv`
 - Description: Elixhauser comorbidities
 - Features in final dataset derived from this dataset:
 - `elix_{elix_name}` – Value of individual Elixhauser category
 - `elix_index_mortality` – Elixhauser Index calculated based on [9]
- `procedure.csv`
 - Description: Procedural history
 - Features in final dataset derived from this dataset:
 - `cpt` – CPT code of the procedure prescribed to the patient
- `cpt.csv`
 - Description: Surgical procedure information
 - Features in final dataset derived from this dataset:
 - `any_surgery` – Prior history of surgical procedures
 - Dependent files:
 - `procedure.csv`
- `col_pressure.csv`
 - Description: Ward-level colonization pressure
 - Features in final dataset derived from this dataset:
 - `{pathogen_name}_cp` – Calculated by the time-weighted 6-month prevalence of `{pathogen_name}` in patients who occupied the same ward as a case or control in the previous 30 days
 - Dependent files:
 - `micro.csv`
 - `ADT.csv`

Data De-identification

- Patient and encounter identifiers were replaced with randomly generated integers to remove all direct identifiers, and `deidentified_patient_id` values were generated independently for the environmental and patient analyses to prevent cross-linkage between datasets.
- Hospital ward was replaced with randomly generated integers.
- To de-identify temporal information in the dataset, each patient was assigned a randomly generated “anchor date” uniformly sampled between January 1, 2000, and January 1, 2100. This anchor date replaced the patient’s earliest recorded date in the dataset. All subsequent dates for the same patient were shifted by the same number of days relative to this anchor date, thereby preserving temporal intervals between events while removing any linkage to real calendar dates.
- For additional privacy protection, patients aged >90 years were recoded into a single category (“>90”).

Usage Notes

Getting Started

To start in R:

Read in the files

```
environmental_df <- read.csv('HO_infxn_environmental_analysis.csv')
```

Calculate the mean C. difficile colonization pressure in cases and controls for different target pathogens

```
cdiff_cp_summary <- environmental_df %>% group_by(run, group) %>% summarize(mean_cp = mean(CDiff_cp, na.rm = TRUE), n = n())

print(cdiff_cp_summary)
```

Source Code for Data Processing and Analysis

All code necessary for building the clean dataset from dependent raw datasets and running the models is available on GitHub [11]. A static snapshot of the code at the time of initial data submission is included in this repository.

Data Usage

This dataset is a de-identified version of the dataset used in [10]. It enables studies of hospital-acquired infection due to infection history of ward co-occupants and prior antibiotic exposures. It was derived from the Mass General Brigham system, and while designed for generalizable research use, results may be influenced by site-specific factors and the matched case-control design.

Release Notes

Version 1.0.0: Initial public release of the dataset.

Ethics

This study was approved by the Institutional Review Board (IRB) of Massachusetts General Brigham healthcare, protocol 2017P000682.

Conflicts of Interest

The authors declare no competing interests.

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